



1 SUMMARY

- The nonlinear and multi-scale nature of plasma turbulence poses a significant challenge for modelling large-scale systems.
- The effect of the small-scale turbulent dynamics into the large-scale energy transfer and transport is a key open question.
- Using Large Eddy Simulations (LES) + Sub-Grid-Scale (SGS) models to address this question in plasmas has been poorly explored [Zhou and Matthaeus, 1990; Yokoi+ 2008; Usmanov+ 2016; Aluie 2017; Chhiber+ 2021].
- The Earth's magnetosheath is numerically and observationally accessible, it can be used as a proxy to validate SGS models and to gain insights to study astrophysical plasmas that are beyond reach.
- We explore the SGS terms in the generalized Ohm's law and how they relate to their filtered counterpart and to the filtered electric and magnetic fields.

2 Filtering the Ohm's law terms

We express any unfiltered quantity as the superposition of filtered and residual terms, i.e., $f = \bar{f} + f'$. We use a Gaussian filter to separate large-scale and to directly measure the anomalous contribution from each term. Applying this procedure to the generalized Ohm's law we get:

$$\bar{\mathbf{E}} = -\bar{\mathbf{u}}_i \times \bar{\mathbf{B}} + \frac{\bar{\mathbf{J}} \times \bar{\mathbf{B}}}{\bar{n}_e} - \frac{1}{\bar{n}_e} \nabla \cdot \bar{\mathbf{P}}_e + \frac{m_e}{\bar{n}_e^2} \left[\nabla \cdot \left(\bar{\mathbf{u}}_i \bar{\mathbf{J}} + \bar{\mathbf{J}} \bar{\mathbf{u}}_i - \frac{\bar{\mathbf{J}} \bar{\mathbf{J}}}{\bar{n}_e} \right) + \frac{1}{\mu_0} \nabla^2 \bar{\mathbf{E}} \right] \quad (1)$$

$$\bar{\mathbf{E}} = -\bar{\mathbf{E}}_{MHD} + \bar{\mathbf{E}}_{Hall} - \bar{\mathbf{E}}_{Pe} + \bar{\mathbf{E}}_{Ine} - \bar{\boldsymbol{\tau}}_{nE} - \bar{\boldsymbol{\tau}}_{nu_i \times B} + \bar{\boldsymbol{\tau}}_{J \times B} + \bar{\boldsymbol{\tau}}_{Ine}, \quad (2)$$

$$\bar{\boldsymbol{\tau}}_{nE} = (\bar{n}\bar{\mathbf{E}} - \bar{n}\bar{\mathbf{E}})/\bar{n}, \quad \bar{\boldsymbol{\tau}}_{nu_i \times B} = (\bar{n}\bar{\mathbf{u}}_i \times \bar{\mathbf{B}} - \bar{n}\bar{\mathbf{u}}_i \times \bar{\mathbf{B}})/\bar{n},$$

$$\bar{\boldsymbol{\tau}}_{J \times B} = (\bar{\mathbf{J}} \times \bar{\mathbf{B}} - \bar{\mathbf{J}} \times \bar{\mathbf{B}})/\bar{n}_e, \quad \bar{\boldsymbol{\Sigma}}\boldsymbol{\tau} = -\bar{\boldsymbol{\tau}}_{nE} - \bar{\boldsymbol{\tau}}_{nu_i \times B} + \bar{\boldsymbol{\tau}}_{J \times B} + \bar{\boldsymbol{\tau}}_{Ine}$$

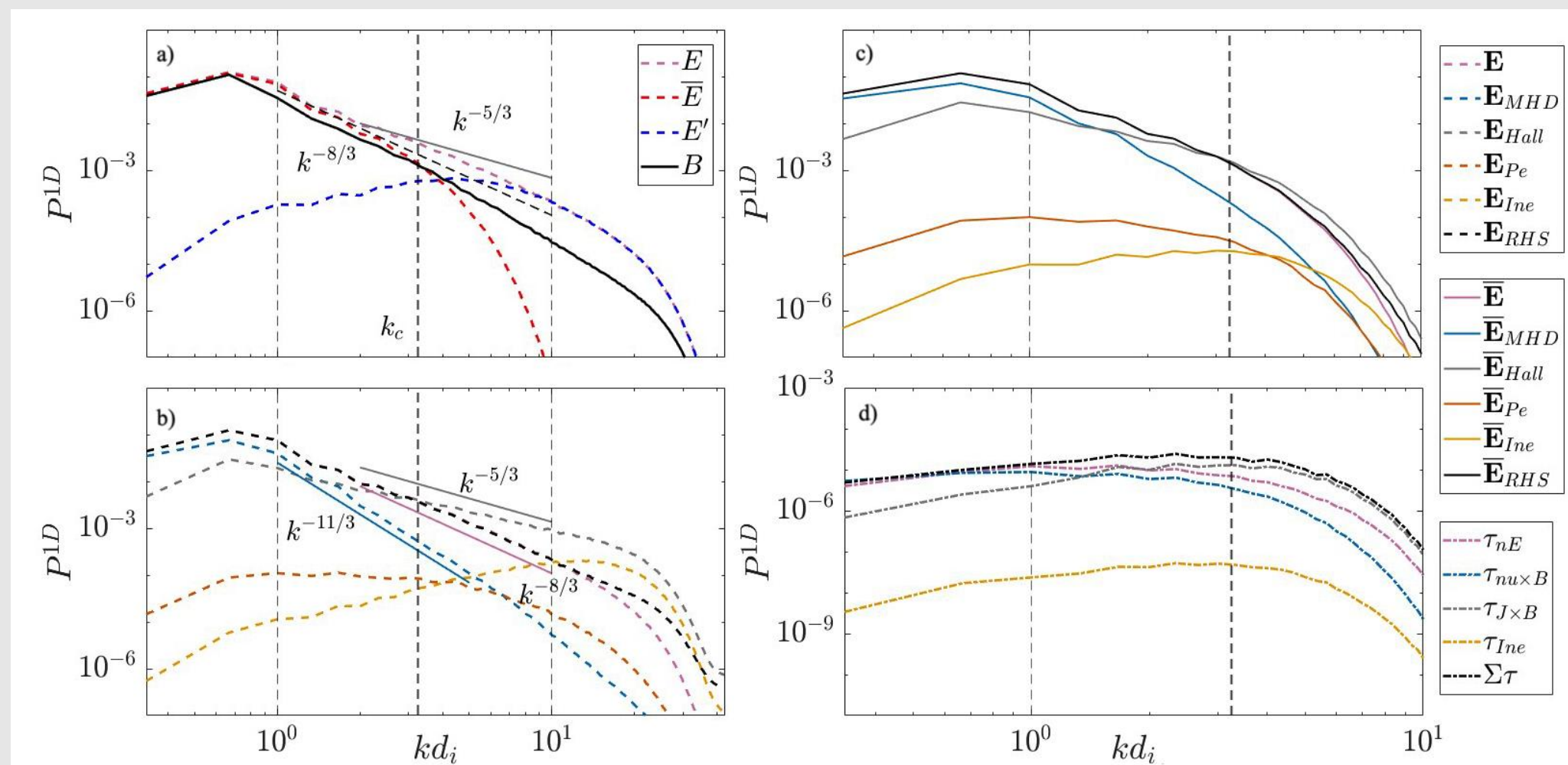


Figure 1. PSD of E contributions from a 3D Vlasov-Hybrid simulation of plasma turbulence in the Earth's magnetosheath [Granier+ 2024]. Spectra of the unfiltered, filtered and anomalous contributions.

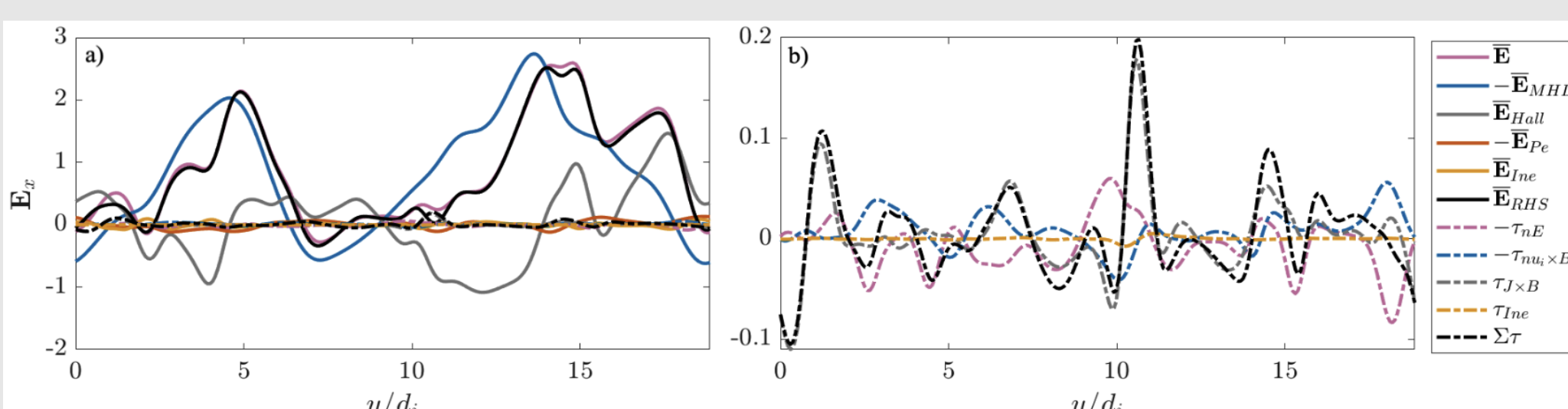


Figure 2. 1D cuts of the resolved and anomalous terms through the simulation domain.

The balance between the anomalous resistivity and anomalous transport is disrupted at regions where the nonlinear correlation at present between v and B .

$$\boldsymbol{\tau}_{nE_{MHD}} = (\bar{n} \bar{\mathbf{u}}_i \times \bar{\mathbf{B}} - \bar{n} \bar{\mathbf{u}}_i \times \bar{\mathbf{B}}) / \bar{n} + \bar{\mathbf{u}}_i \times \bar{\mathbf{B}} - \bar{\mathbf{u}}_i \times \bar{\mathbf{B}} \quad (3)$$

$$\boldsymbol{\tau}_{nE_{MHD}} = -\boldsymbol{\tau}_{nu_i \times B} + \boldsymbol{\tau}_{u_i \times B},$$

$$\boldsymbol{\tau}_{nE_{EMHD}} = -\boldsymbol{\tau}_{nu_e \times B} + \boldsymbol{\tau}_{u_e \times B}, \quad (4)$$

At each regime a balance between anomalous resistivity and other anomalous terms can be present. Are the nonlinearities with density fluctuations really important for the reconnection electric field?

3 Characterising contributions

Figure 3. (a-b) Spectral ratios and phase-coherence between anomalous and filtered terms. (c-d) Phase-coherence, contribution to the generation of filtered E and B.

Take away points:

- The anomalous terms preserve partial coherence with their filtered counterpart.
- The anomalous terms mainly oppose (partially anti-correlated) filtered E.
- The solenoidal part of the anomalous terms show a diamagnetic effect by opposing the generation of filtered magnetic field.

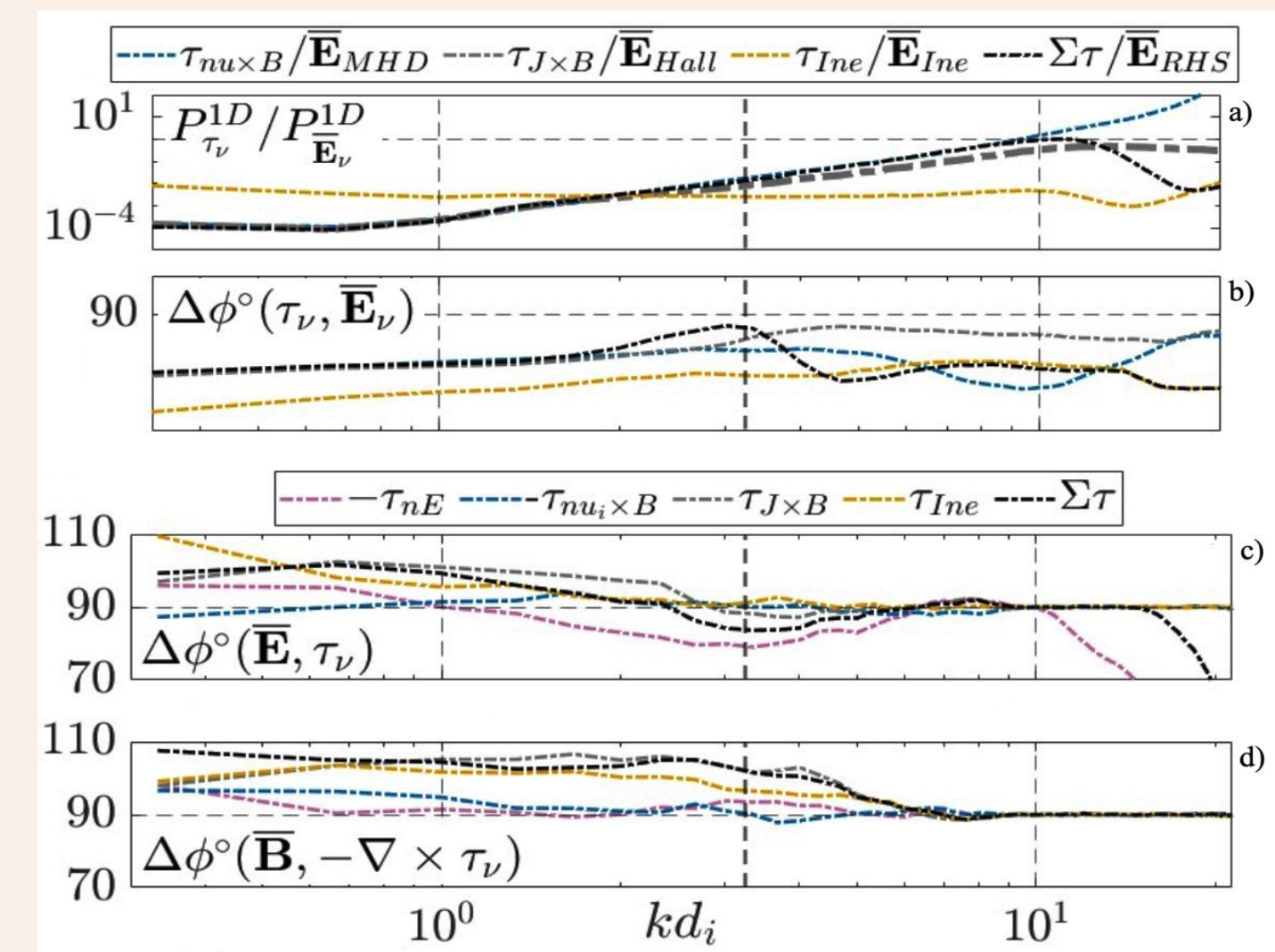
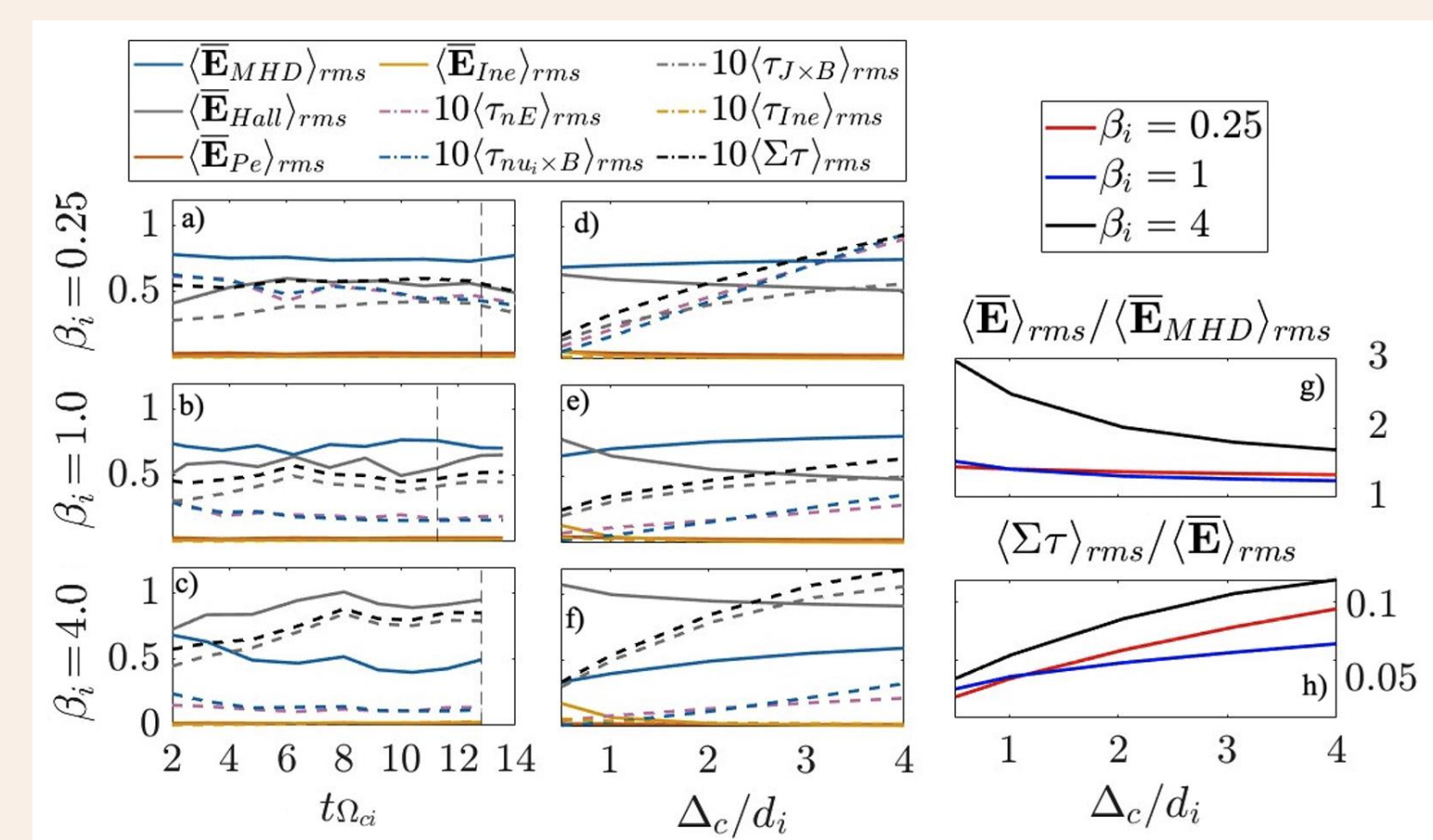


Figure 4. Temporal evolution of the ratios between anomalous terms and dependence with filter scale for three ion plasma betas.



- The contribution from the Hall electric field increases as the time progresses. This is clearer for the large ion plasma beta case.
- The increase in the anomalous electric field for larger filter scale involves the phase coherence.

4 Testing SGS models

At MHD level, the unresolved Reynolds stress (R) and the unresolved electromotive force (EM) are modelled in Eq. (5-6).

$$\frac{\partial \bar{\mathbf{u}}}{\partial t} + \nabla \cdot (\bar{\mathbf{u}} \bar{\mathbf{u}} - \bar{\mathbf{B}} \bar{\mathbf{B}}) = -\nabla P_M + \nabla \cdot \bar{\mathbf{R}} + \nu \nabla^2 \bar{\mathbf{u}}, \quad (5)$$

$$\frac{\partial \bar{\mathbf{B}}}{\partial t} = \nabla \times (\bar{\mathbf{u}} \times \bar{\mathbf{B}} + \bar{\mathbf{E}}_M) + \lambda \nabla^2 \bar{\mathbf{B}} \quad (6)$$

$$\Omega = \nabla \times \bar{\mathbf{u}}; \quad W = \bar{\mathbf{u}}' \cdot \bar{\mathbf{b}}'; \quad E_M = \bar{\mathbf{u}}' \times \bar{\mathbf{B}}' = -\beta \bar{\mathbf{J}} + \alpha \bar{\mathbf{B}} + \gamma \bar{\Omega}, \quad (7)$$

$$W = \bar{\mathbf{u}}' \cdot \bar{\mathbf{b}}', \quad H = -\bar{\mathbf{u}}' \cdot \bar{\Omega}' + \bar{\mathbf{b}}' \cdot \bar{\mathbf{J}}', \quad K = \frac{\bar{\mathbf{u}}'^2 + \bar{\mathbf{b}}'^2}{2}, \quad KR = \frac{\bar{\mathbf{u}}'^2 - \bar{\mathbf{b}}'^2}{2} \quad (8)$$

$$\begin{aligned} \Sigma \tau_{M1} &= C_\alpha \bar{\mathbf{B}} + C_\beta \bar{\mathbf{J}} + C_\gamma \bar{\Omega}, \\ \Sigma \tau_{M2} &= C_\alpha H \bar{\mathbf{B}} + C_\beta (K - KR) \bar{\mathbf{J}} + C_\gamma W \bar{\Omega}, \\ \Sigma \tau_{MHD} &= \bar{\mathbf{E}}_{Hall} - \bar{\mathbf{E}}_P + \bar{\mathbf{E}}_{Ine} + \Sigma \tau, \\ \Sigma \tau_{Hall} &= -\bar{\mathbf{E}}_P + \bar{\mathbf{E}}_{Ine} + \Sigma \tau, \end{aligned} \quad (9-12)$$

Figure 5. a) Spectrum of the models for the anomalous and effective-anomalous and effective. **b)** Phase coherence between models and measured anomalous.

b) Phase coherence between models and measured anomalous.

Take away points:

- These models (Eqs. 9-12) do not capture the coherence information from the measured total anomalous term.
- A structural approach, seems to be required for SGS models in collisionless plasmas.

Future work:

- Comparison with measurements from turbulent intervals in the Earth's magnetosheath using the Magnetospheric multiscale (MMS) mission.
- Test semi empirical and structural models that provide information about coherence.

